

Supporting Information

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Ionic Conductive Tunnels in *h*-WO₃ Enable High-Rate NH₄⁺ Storage

Yi-Zhou Zhang, Jin Liang, Zihao Huang, Qian Wang, Guoyin Zhu, Shengyang Dong*, Hanfeng Liang* and Xiaochen Dong*



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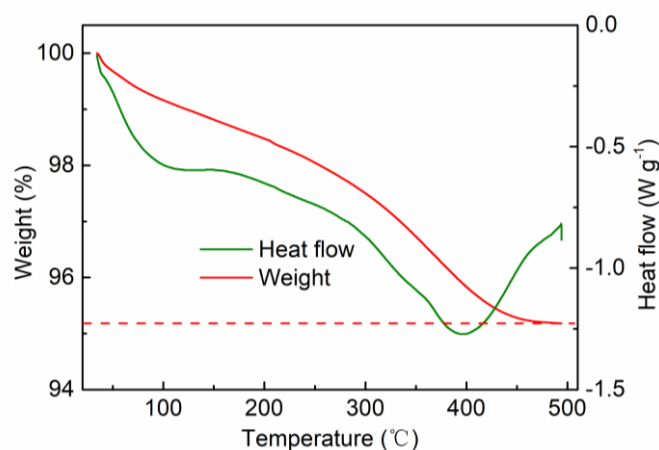


Figure S1. TG/DSC profile of h -WO₃ powder.

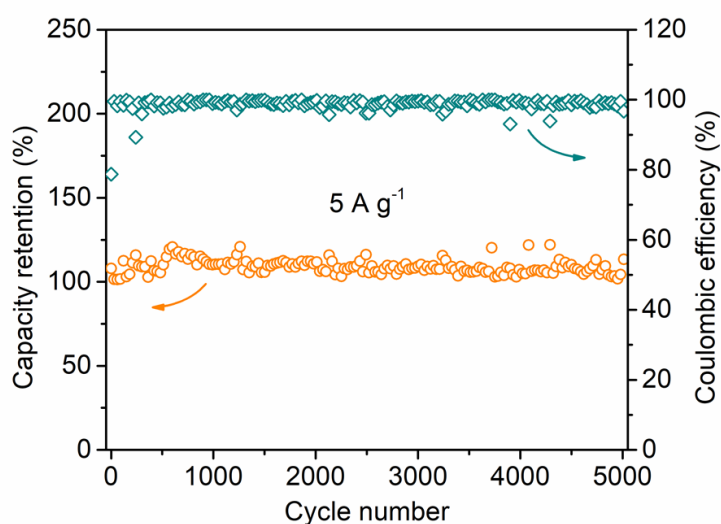


Figure S2. Cycling performance of h -WO₃ in 1 M (NH₄)₂SO₄ at 5 A g⁻¹.

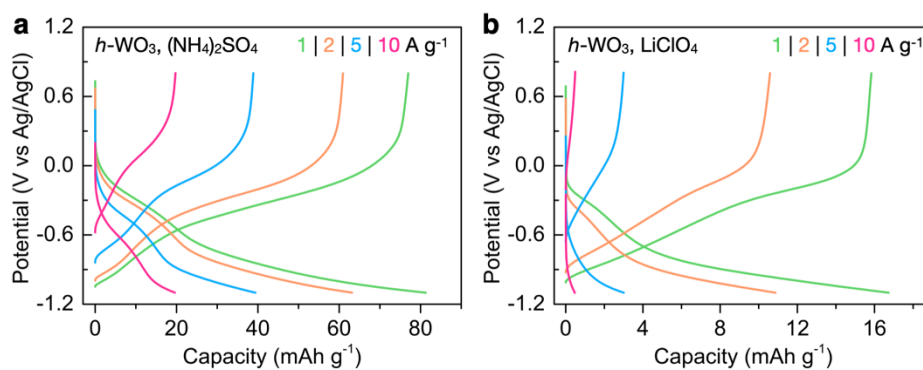


Figure S3. The charge-discharge curves of $h\text{-WO}_3$ electrode with 1 M $(\text{NH}_4)_2\text{SO}_4$ (a) and LiClO_4 (b) as the electrolyte under various current densities.

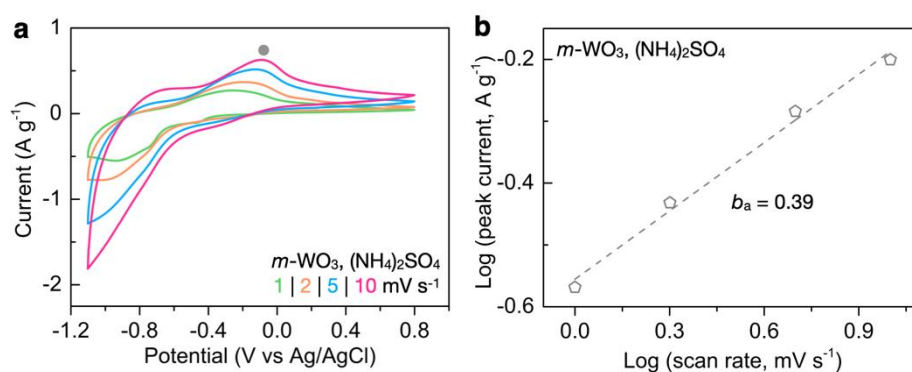


Figure S4. (a) CV curves at different scan rates and (b) the b value of $m\text{-WO}_3$ in 1 M $(\text{NH}_4)_2\text{SO}_4$ electrolyte.

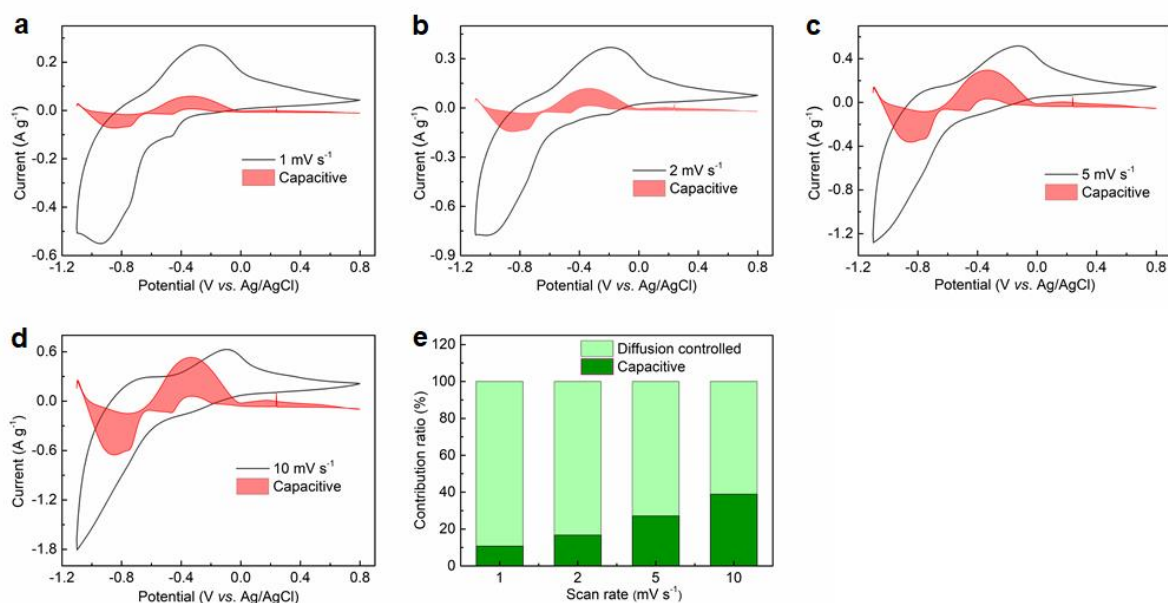


Figure S5. Separation of the capacitive and diffusion currents of *m*-WO₃ at (a) 1 mV s⁻¹, (b) 2 mV s⁻¹, (c) 5 mV s⁻¹, (d) 10 mV s⁻¹. (e) Contribution of the capacitive behavior of 1 (10.7%), 2 (16.7%), 5 (21.1%) and 10 (38.8%) mV s⁻¹.

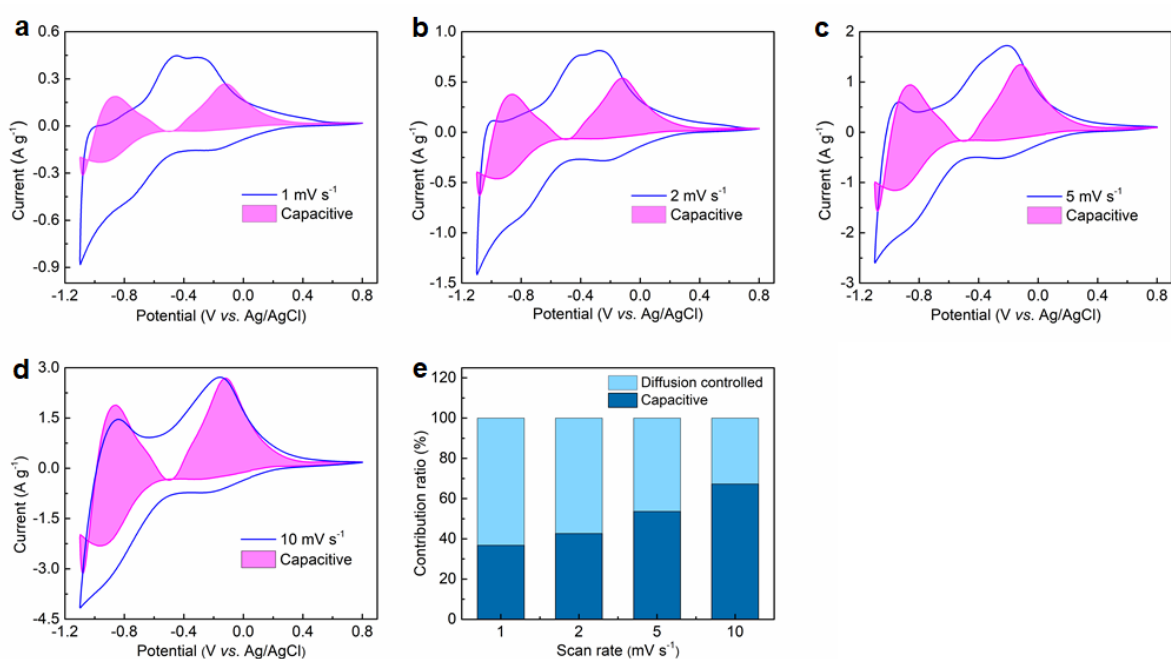


Figure S6. Separation of the capacitive and diffusion currents of *h*-WO₃ at (a) 1 mV s⁻¹, (b) 2 mV s⁻¹, (c) 5 mV s⁻¹, (d) 10 mV s⁻¹. (e) Contribution of the capacitive behavior of 1 (36.8%), 2 (42.7%), 5 (53.6%) and 10 (67.3%) mV s⁻¹.

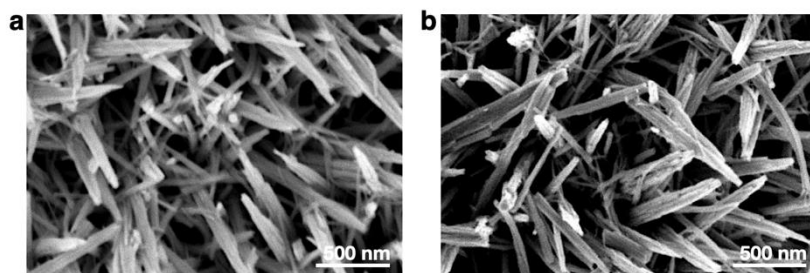


Figure S7. SEM images of the $h\text{-WO}_3$ after (a) discharge and (b) charge.

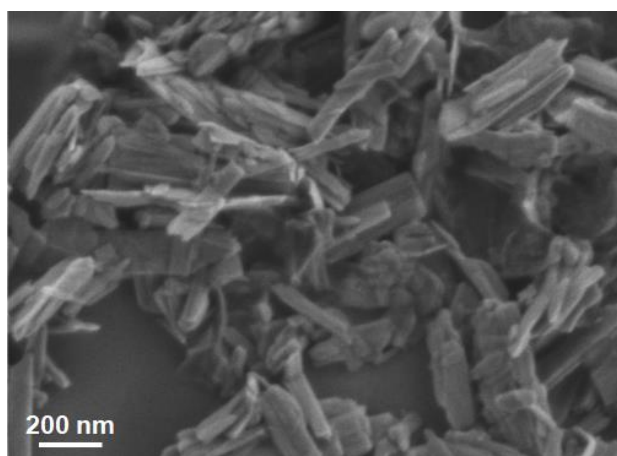


Figure S8. SEM image of $(\text{NH}_4)_{0.5}\text{V}_2\text{O}_5$.

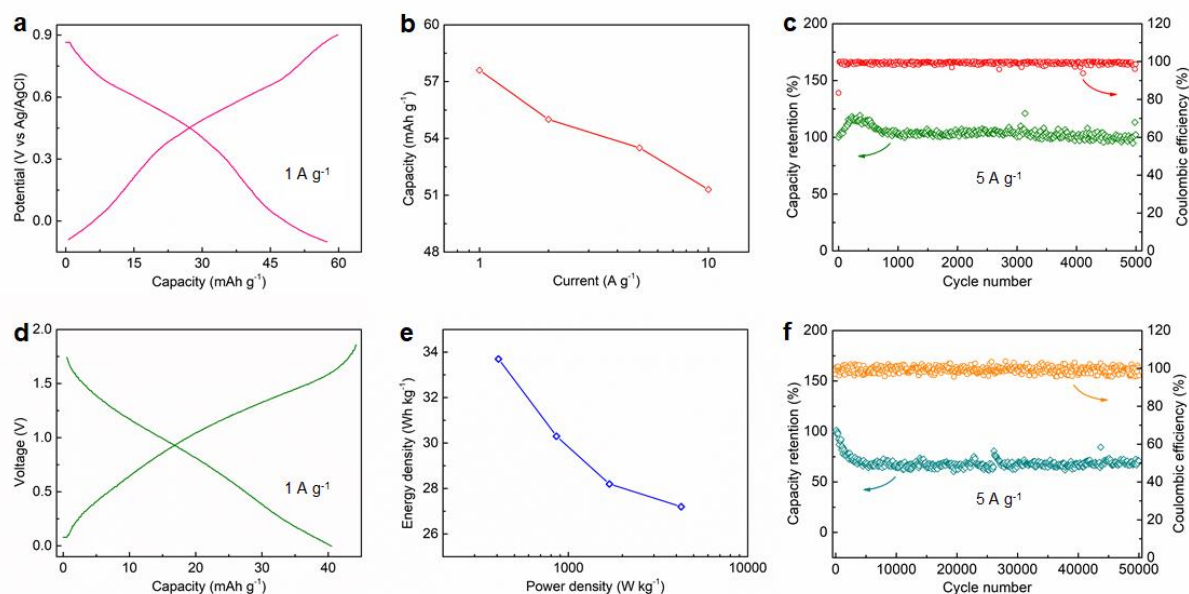


Figure S9. Electrochemical performance of $(\text{NH}_4)_{0.5}\text{V}_2\text{O}_5$ nanobelts. (a) Typical CD curve at a current density of 1 A g^{-1} . (b) Rate capacities. (c) Cycling stability at 5 A g^{-1} . Electrochemical performance of $(\text{NH}_4)_{0.5}\text{V}_2\text{O}_5/\text{h-WO}_3$ full cell. (d) Typical CD curve at a current density of 1 A g^{-1} . (e) Ragone curves. (f) Cycling stability at 5 A g^{-1} .

Table S1. Calculated energies (eV) for the h-WO_3 and NH_4^+ intercalated h-WO_3 .

E_{initial}	E_{ammonium}	E_{hybrid}	ΔE
-472.60	-17.61	-530.45	-11.32

The values listed above (from left to right) present the energies of the initial state (pristine h-WO_3 without NH_4^+ intercalation), one NH_4^+ ion, and hybrid state (h-WO_3 with two intercalated NH_4^+ ions), respectively. The variation of intercalation energy caused by one NH_4^+ insertion (ΔE) was calculated according to $\Delta E = [E_{\text{hybrid}} - (2 E_{\text{ammonium}} + E_{\text{initial}})]/2$, whereas the average guest-host interaction energy for NH_4 (4 bonds) per bonding was $\Delta E/4$.

Table S2. The charge and charge transfer of two NH_4^+ ions upon insertion into $h\text{-WO}_3$.

Atom	Charge	Charge transfer
N	6.299876	-0.700124
H	0.461923	-0.538077
H	0.429611	-0.570389
H	0.454021	-0.545979
H	0.478277	-0.521723
N	6.300366	-0.699634
H	0.461724	-0.538276
H	0.429622	-0.570378
H	0.453892	-0.546108
H	0.478061	-0.521939
Per NH_4^+		-2.876314